Thermo-Credit Theory: A Credit-First Thermodynamic Mapping

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Abstract

This note presents Thermo Credit as a measurement-first macro-credit framework. The central question is simple: when new credit goes to different destinations, how does that change growth, inflation, asset-price pressure, and fragility? The model has two parts. The first is a stock-flow-consistent accounting base built from balance-sheet identities. The second is a credit state layer that tracks capacity, stress, persistence, and adjustment costs. We use a compact state built around credit partition, capacity, and stress, including a bookkeeping potential $U(S_{\mathrm{M}}, V_{\mathrm{C}}, \xi, \ldots)$ with memory variable ξ. We also define practical gauges such as free-energy-like F_{C} and exergy-like X_{C}. A Maxwell-like condition is used as a local consistency check, not as a general law. When it fails, we read the gap as observable non-equilibrium behaviour in the $(S_{\mathrm{M}}, V_{\mathrm{C}})$ plane. The aim is practical: the quantities should be computable from public data, tested by forecast value, and released in a form that others can recalculate. In this revision, region-specific reference parameters are treated as tunable values rather than fixed constants, and the note states the implicit relation and the calibration objective more directly. The credit split is also sharpened: the theory now distinguishes GDP-linked credit, construction credit, and credit used to buy existing assets such as land, housing, and financial claims.

1 Problem Statement and Scope

Classical money-first views such as the Quantity Theory of Money (QTM) explain nominal dynamics through the money stock, its velocity, and policy or shock terms [1, 2, 3]. Credit-first views, notably the Quantity Theory of Credit (QTC), stress that banks create deposits when they lend and that the use of credit matters for macro-financial outcomes [4, 5]. In parallel, information theory and statistical mechanics have inspired useful analogies for allocation and dispersion [6, 7, 8, 9, 10].

This note takes a limited and practical step. We build a thermodynamic-style state description that separates scale from dispersion, adds an explicit capacity variable for bank balance sheets, and yields testable constraints and early-warning gauges. The claim is not that macro data follow physical laws. The claim is narrower: a disciplined bookkeeping analogy can help supervision, stress testing[K18], and systematic monitoring.

Research target and use cases

The measurement-first revision narrows the framework to one question: how does the destination of new credit change the path from credit expansion to growth, inflation, asset-price acceleration, and fragility? In practical terms, the current version serves three use cases:

1. early-warning signals for asset inflation and systemic fragility;
2. description of the GDP-linked versus existing-asset credit mix rather than credit volume alone;

3. forecast improvement relative to total-credit-only baselines.

This keeps the theory tied to a concrete mapping from inputs to outputs. It also stops the thermodynamic language from turning into a loose metaphor.

Related Work

Several research strands motivate this mapping. Market microstructure work has already proposed temperature- and entropy-like summaries of liquidity, but it mainly stays at the order-book level and does not address credit allocation or regulatory headroom [12]. Work on model risk also uses thermodynamic-style language, but not in a bank balance-sheet setting [13].

This note works at a coarser scale. It builds (S_M, V_C, T_L, p_C) from balance-sheet aggregates, credit allocation shares, and market-stress proxies. The aim is not to reproduce thermodynamics in full. It is to create a compact macro-credit state description that can be tested against growth, spread, and fragility outcomes.

2 Credit Partition and State Variables

We work over a chosen system (jurisdiction, sector set) and time aggregation (e.g. monthly). Let M_{in} denote money-in-circulation[K10] over this domain, and let $\{q_i\}$ be a stable, MECE (Mutually Exclusive, Collectively Exhaustive)[K15] partition of its uses (real activity, housing credit, securities margin, etc.), with

$$\sum_i q_i = 1.$$

2.1 Measurement-first overlay: credit partition and targets

To make the framework operational, we add a minimal measurement-first overlay. Its purpose is narrower than the full thermodynamic analogy. The working question is: how does the destination of new credit change the translation of credit expansion into output, prices, asset prices, and fragility? In that sense, the overlay is close to Werner's destination-of-credit view: the main issue is not credit in the abstract, but whether new credit funds GDP-linked transactions or mainly bids up existing assets.

The smallest useful state vector is

$$\left(C_t, q_t, Y_t^N, Y_t^R, P_t, A_t, S_t\right),$$

where C_t denotes new credit flow during period t, $q_t \in [0,1]$ is the share of that flow directed to GDP-linked transactions, Y_t^N is a nominal output proxy used for accounting ratios, Y_t^R is a real-activity target, P_t is a broad price-level proxy, A_t is an asset-price or balance-sheet accumulation proxy for existing assets, and S_t is a stress / fragility state. The coarse GDP-linked-versus-existing-asset split is then

$$C_t^R = q_t C_t, \qquad C_t^A = (1 - q_t) C_t. \tag{1}$$

For empirical work, however, the two-way split is often still too coarse. Following the destination-of-credit logic, a more operational decomposition is

$$C_t = C_t^G + C_t^B + C_t^E, \tag{2}$$

where C_t^G funds GDP-linked transactions such as working capital and business investment, C_t^B funds construction and development, and C_t^E funds purchases of existing assets such as land, existing housing, or financial claims. The two-way quantities in (1) can then be recovered as

$$C_t^R = C_t^G + \lambda_B C_t^B, \qquad C_t^A = C_t^E + (1-\lambda_B) C_t^B, \qquad q_t = \frac{C_t^R}{C_t}, \tag{3}$$

with $\lambda_B \in [0,1]$ representing the share of construction credit that behaves like current output formation rather than like collateral-driven asset inflation. This keeps the theory simple at the top level while allowing the measurement layer to treat construction as a genuine intermediate case.

This partition share q_t should not be confused with the entropy shares $\{q_i\}$ used below. The scalar q_t classifies *new credit flows* into GDP-linked and existing-asset destinations. By contrast, $\{q_i\}$ partitions broader money and credit uses for the entropy construction. The two objects are related, but they are not the same.

The labels should also be read with care. In the present note, C_t^R is short for credit that is closely linked to current output, income, or new production. C_t^A is short for credit that mainly finances purchases of existing land, existing housing, or financial assets. Construction credit sits between these two cases: it can add to current GDP, but it can also feed collateral and land-price cycles. In applied work it may therefore need a partial weight rather than a hard bucket.

Identification of q_t is best handled in layers: (i) direct observation from sectoral lending and loan-purpose data when available, (ii) proxy inference from housing, land, equity, investment, and stress variables when direct splits are incomplete, and (iii) latent-state estimation only after the reduced-form version is stable.

The practical mapping should follow loan purpose rather than borrower labels alone. Credit to non-financial firms can still be asset-directed if it mainly funds land acquisition or leveraged purchases of existing property. Conversely, some real-estate credit is GDP-linked when it finances new building activity. This is why the loan-purpose map in (2) is more useful than a simple household-versus-firm split.

2.2 Monetary dispersion entropy

We define an entropy-like extensive index

$$S_{\mathrm{M}} = k M_{\mathrm{in}} H(q), \qquad H(q) \equiv -\sum_i q_i \log q_i, \tag{4}$$

where $k > 0$ is a conventional scaling constant. This mirrors ideal mixing entropy $\Delta S_{\mathrm{mix}} \propto N H(x)$ and follows the information-theoretic form of Shannon [9]. S_{M} grows with both scale and dispersion; concentration (reallocation into fewer uses) reduces $H(q)$ and can reduce S_{M} even at fixed M_{in}.

Formally, $H(q)$ is the Shannon–Boltzmann entropy of the MECE bucket probabilities $\{q_i\}$ with natural logarithm (no Tsallis, Rényi, or alternative families are used here). In addition to the extensive S_{M}, the implementation tracks a scale-free version $S_{\mathrm{M,hat}} = H(q)/\log K$ (with K buckets) and per-bucket contributions $S_{\mathrm{M,in}}^{(i)} = k M_{\mathrm{in}} \tilde{q}_i$. For the baseline JP configuration these buckets correspond to "productive", "housing", "consumption", "financial", and "government" allocations (recorded as $q_{\mathrm{productive}}, q_{\mathrm{housing}}, q_{\mathrm{consumption}}, q_{\mathrm{financial}}, q_{\mathrm{government}}$); regional variants use analogous coarse splits in their allocation tables, always normalised so that $\sum_i q_i = 1$.

Energy and entropy conventions. The internal potential $U(S_{\mathrm{M}}, V_{\mathrm{C}}, \ldots)$ is a bookkeeping scalar in currency units, and the derived quantities F_{C}, X_{C}, and G_{C} are free-energy- and exergy-like gauges

constructed from $(U, S_{\mathrm{M}}, V_{\mathrm{C}})$ and fixed environmental parameters (T_0, p_0). They are interpreted as indicators of usable credit capacity and policy room, not as literal physical energies; general caveats on the analogy are collected in Section 5.

2.3 Capacity and conjugate variables

We introduce:

- V_{C}: an effective credit capacity or headroom[K4] (e.g. CET1/RWA[K3] slack, HQLA (High Quality Liquid Assets)[K6]-based lending space, liquidity buffers[K7]);
- T_{L}: a liquidity-intensity index (a "temperature-like" proxy from spreads, turnover, depth);
- p_{C}: a "credit pressure"[K5] (shadow price of capacity).

We use a quasi-potential (bookkeeping energy) with an explicit memory variable ξ that captures slow-moving structural distortions (regime, collateral and haircut conventions, concentration, funding fragility, etc.):

$$U = U(S_{\mathrm{M}}, V_{\mathrm{C}}, \xi, \ldots). \tag{5}$$

We then define reference conjugate quantities

$$T_{\mathrm{L}} \equiv \left(\frac{\partial U}{\partial S_{\mathrm{M}}}\right)_{V_{\mathrm{C}},\xi}, \qquad p_{\mathrm{C}} \equiv -\left(\frac{\partial U}{\partial V_{\mathrm{C}}}\right)_{S_{\mathrm{M}},\xi}. \tag{6}$$

Here U is not a physical internal energy; it is a bookkeeping potential chosen so that a first-law-like decomposition holds. Classical QTM/QTC do not name U; we introduce it as an analytic device. At the level of the abstract mapping, (6) should be read as a reference structure: in an idealized state description there exists some smooth potential $U_{\mathrm{th}}(S_M, V_C)$ whose canonical fields $(T_L^{\mathrm{th}}, p_C^{\mathrm{th}})$ are given by these derivatives. The empirical objects $U(t)$, $T_L(t)$, and $p_C(t)$ used in this note are proxy series constructed from GDP-like aggregates, liquidity metrics, and regulatory headroom; they are chosen to be interpretable and to track the intended directions, but they are not literally defined by finite-difference approximations to (6) in the code.

Non-equilibrium entropy balance. In an open macro-financial system, a monotonic "second law" for S_{M} is not appropriate. Instead we work with an entropy-like balance

$$\frac{\mathrm{d}S_{\mathrm{M}}}{\mathrm{d}t} = \Phi_S(t) + \sigma_S(t), \qquad \sigma_S(t) \geq 0, \tag{7}$$

where Φ_S is an entropy flow induced by external coupling (policy, foreign flows, institutional shifts) and σ_S is a non-negative entropy production rate associated with irreversible credit reallocation and market fragmentation. In this note we treat σ_S operationally and proxy it using hysteresis and loop-area diagnostics (Section 4); S_{M} itself may rise or fall depending on the sign of Φ_S.

Empirically, $U(t)$ is instantiated as a scalar credit-capacity gauge constructed from GDP-like or credit aggregates. When a dedicated U series is not available, the implementation fills it using a conservative fallback order over available GDP-like and credit-stock series (see Appendix, Internal-energy gauge). An optional detrended version $U^\star$ is used only for diagnostics and visualisation (to study fluctuations around a slowly varying baseline), while the core potentials F_{C} and X_{C} are computed from the undetrended U.

2.4 Operational metrics and irreversibility

The v2 overlay turns the thermodynamic language into directly computable operators. A minimal set of empirical ratios is

$$\eta_t \equiv \frac{\Delta Y_t^N}{C_t}, \tag{8}$$

$$d_t \equiv \frac{\Delta A_t}{C_t}, \tag{9}$$

$$\sigma_t \equiv \max\{0, 1 - \eta_t\}, \tag{10}$$

where η_t is a credit-efficiency proxy, d_t is an asset-bias proxy, and σ_t is a deliberately simple dissipation proxy. When only an asset-price index is available, a normalized alternative is

$$d_t^{\text{proxy}} \equiv \frac{\Delta \log A_t}{C_t / Y_t^N}. \tag{11}$$

To make the "Thermo" label empirically useful, we also need a minimal irreversibility block. The reduced-form version used in the current v2 draft is

$$\eta_t = \eta_0 - \alpha(1 - q_t) - \beta S_{t-1} + \varepsilon_{\eta,t}, \tag{12}$$

$$S_t = \rho S_{t-1} + \gamma \frac{C_t^A}{Y_t^N} - \delta \frac{C_t^R}{Y_t^N} + \varepsilon_{S,t}, \tag{13}$$

so that credit flowing into existing-asset transactions raises fragility while accumulated stress lowers next-period conversion efficiency.

Under the three-way partition in (2), the same idea can be written more explicitly by replacing C_t^A with $C_t^E + (1 - \lambda_B) C_t^B$ and replacing C_t^R with $C_t^G + \lambda_B C_t^B$. The stress block is therefore meant to react most strongly to credit that supports prices of existing assets, not to all property-related credit indiscriminately.

These expressions are not presented as a full structural law. They are a measurement-first bridge from theory to data. Existing diagnostics such as S_{M}, T_{L}, p_{C}, loop area, U, F_{C}, and X_{C} should therefore be read as companion indicators and possible inputs into the stress block S_t, not as substitutes for the partition variables (C_t^R, C_t^A, q_t).

2.5 Depth and turnover proxies

To construct the liquidity temperature T_{L} and related gauges from observable series, we introduce two derived inputs: a *depth* proxy D and a *turnover* proxy Θ. They are practical inputs rather than state variables in their own right. Depth is usually a rescaled version of L_{real}, while turnover is usually a safe version of U/L_{real}. The implementation clips or fills pathological cases so that one noisy observation does not dominate the temperature measure.

Design of the liquidity temperature. The liquidity temperature is designed to be high when spreads are narrow, markets are deep, and turnover is brisk. It should fall sharply when any one of these ingredients becomes stressed. To this end we combine standardized components (z_s, z_d, z_t)—the z-scores of inverse spreads, inverse depth, and turnover—into a composite

$$\hat{T}_{\text{mult}}(t) = z_s(t)\, z_d(t)\, \left(1 + \tfrac{1}{2} z_t(t)\right),$$

and then rescale to $T_{\mathrm{L}} \in [0, 1]$ by min–max normalisation. We use this multiplicative form as the baseline because it encodes a simple "weakest link" logic: severe stress in one component should pull the whole temperature measure down.

Table 1: Thermodynamic–QTM–QTC correspondence (analogy-level).

Variable	Thermodynamics	QTM (money-first)	QTC (credit-first)
Mixing entropy	ΔS_{mix}	$S_{\mathrm{M}} = kM_{\mathrm{in}}H(q)$	$S_{\mathrm{M}} = kM_{\mathrm{in}}H(q)$
Temperature	T	T_L (proxy)	T_{L} (liquidity proxy)
Internal energy	U	$U_{\mathrm{M}}(S_{\mathrm{M}})$ (auxiliary)	$U(S_{\mathrm{M}}, V_{\mathrm{C}}, \ldots)$
Volume	V	–	V_{C} (capacity/headroom)
Pressure	p	–	$p_{\mathrm{C}} = -(\partial U/\partial V_{\mathrm{C}})_{S_{\mathrm{M}}}$
Heat-like term	$\delta Q_{\mathrm{rev}} = T\,\mathrm{d}S$	$Q_{\mathrm{M}} \sim T_0 \Delta S_{\mathrm{M}}$	$Q_{\mathrm{C}} \sim \bar{T}_L\, \Delta S_{\mathrm{M}}$
Work-like term	$\delta W = p\,\mathrm{d}V$	$W_{\mathrm{M}} \equiv W_{\mathrm{policy}}$	$W_{\mathrm{C}} \equiv -\bar{p}_{\mathrm{C}} \Delta V_{\mathrm{C}} + W_{\mathrm{policy}}$
First law	$\Delta U = T\Delta S + W$	$\Delta U_{\mathrm{M}} = T_0 \Delta S_{\mathrm{M}} + W + \varepsilon$	$\Delta U = \bar{T}_L \Delta S_{\mathrm{M}} + W + \varepsilon$
Second-law tendency	$\Delta S \geq 0$[1]	$\dot{S}_{\mathrm{M}} = \Phi_S + \sigma_S,\ \sigma_S \geq 0$	$\dot{S}_{\mathrm{M}} = \Phi_S + \sigma_S,\ \sigma_S \geq 0$
Helmholtz free energy	$F = U - TS$	$F_{\mathrm{M}} \equiv U_{\mathrm{M}} - T_0 S_{\mathrm{M}}$	$F_{\mathrm{C}} \equiv U - T_0 S_{\mathrm{M}}$
Exergy/availability	$X \approx U - T_0 S$	$X \approx U_{\mathrm{M}} - T_0 S_{\mathrm{M}}$	$X_{\mathrm{C}} = \Delta U + p_0 \Delta V_{\mathrm{C}} - T_0 \Delta S_{\mathrm{M}}$

2.6 Units and scaling conventions

The units are consistent but conventional. M_{in}, L_{real}, U, and V_{C} are monetary or currency-like aggregates; T_{L} and p_{C} are dimensionless indices; and $S_M = kM_{\mathrm{in}}H(q)$ scales with money volume and dispersion. In applications, signs, changes, and relative levels matter more than absolute magnitudes. That is why the empirical section reports a tuned implicit headroom score rather than treating raw nominal X_C levels as directly comparable across regions.

3 Thermodynamic Interpretation

3.1 Correspondence (QTM vs. QTC)

Table 1 summarizes the analogy. Thermodynamic quantities are literal on the left column; QTM/QTC entries are analogy-level constructs. Heat- and work-like pieces are defined for bookkeeping only.

The construction is chosen so that: (i) S_{M} is extensive and sensitive to dispersion; (ii) V_{C} carries explicit capacity/headroom information; (iii) QTC—unlike classic QTM—naturally admits a pressure-like channel p_{C}.

3.2 SFC conservation backbone (accounting first law)

For early-warning use, we anchor the framework in stock–flow consistent (SFC) accounting. This is the non-negotiable conservation layer. It is not an analogy, but a set of identities implied by double-entry bookkeeping and flow-of-funds accounts.

Sectoral financial balances. Let $B_s(t)$ denote the net lending/borrowing (financial balance) of sector s over a period t (positive for net lenders, negative for net borrowers). Using a minimal, data-friendly three-sector aggregation $s \in \{P, G, R\}$ (domestic private, government, rest of world), SFC implies

$$B_P(t) + B_G(t) + B_R(t) = 0. \tag{14}$$

In practice B_G is available as the general-government net lending/borrowing (balance), and B_R is closely related to the current account (up to sign and conventions). When a direct private-sector

[1] The textbook thermodynamic statement refers to a closed irreversible system. In this note the macro-financial system is open, so the operational form is the entropy-balance expression in (7).

net lending series is unavailable, B_P can be computed residually from (14), which is often the most robust route in public-data pipelines.

Stock–flow identity. Let $A_s(t)$ be a stock variable associated with sector s (e.g., net financial assets or a balance-sheet aggregate available at the chosen frequency). The SFC update is

$$A_s(t) = A_s(t-1) + B_s(t) + \mathrm{VAL}_s(t), \tag{15}$$

where $\mathrm{VAL}_s(t)$ collects valuation/reclassification effects. For monitoring, one can either (i) work with series where valuation is small at the aggregation level, (ii) treat VAL_s as an explicit residual, or (iii) use detrended / change-based diagnostics that are less sensitive to level revisions.

Interpretation inside Thermo-Credit. Equations (14)–(15) provide the hard constraint layer. The "energy-like" decomposition that follows is a structured way to attribute observed movements to dispersion (S_M), capacity (V_C), policy work, and non-equilibrium losses, while still respecting the SFC backbone.

When a bank grants a new loan, it simultaneously creates a matching deposit. On the joint bank–customer system, this is a pure accounting operation ("loans create deposits"), consistent with the SFC backbone in Section 3.2. Within our mapping, the associated change in the potential U is decomposed as

$$\Delta U \approx \bar{T}_L\, \Delta S_\mathrm{M} - \bar{p}_\mathrm{C}\, \Delta V_\mathrm{C} + \bar{A}\, \Delta\xi + W_\mathrm{policy} - \mathcal{D} + \varepsilon, \tag{16}$$

where:

- ΔS_M: change in dispersion entropy from scale and allocation shifts (typically $\Delta S_\mathrm{M} > 0$ in net expansion);
- ΔV_C: change in effective capacity ($\Delta V_\mathrm{C} < 0$ when headroom is used up; then $-\bar{p}_\mathrm{C}\Delta V_\mathrm{C} > 0$);
- $\Delta\xi$: change in a memory / distortion variable capturing slow structural shifts (regime, collateral conventions, concentration, funding fragility). $\bar{A}$ is its conjugate "affinity".
- $\mathcal{D} \geq 0$: a dissipation term capturing irreversible loss of usable credit capacity due to hysteresis, fragmentation, and forced balance-sheet adjustments.
- W_policy: structured policy work[K14] from regulation, guarantees, central bank operations, and other deliberate interventions;
- ε: residual term capturing model error and omitted channels.

Credit creation phases (new lending > repayments) and contraction phases (repayments > new lending) then exhibit characteristic sign patterns for $(\Delta M_\mathrm{in}, \Delta S_\mathrm{M}, \Delta V_\mathrm{C}, -\bar{p}_\mathrm{C}\Delta V_\mathrm{C})$, which can be tabulated and compared with data. The value of (16) is diagnostic. It forces us to attribute changes to dispersion, capacity use, or policy work[K14] instead of conflating them.

For intuition it is useful to summarize typical sign patterns. Table 2 collects a minimal classification in terms of net expansion/contraction and whether headroom is being used or rebuilt; the qualitative readings in the caption are meant as guides rather than axioms. These patterns provide a qualitative checklist against which observed $(\Delta M_\mathrm{in}, \Delta S_\mathrm{M}, \Delta V_\mathrm{C})$ can be compared, and can be refined or extended in empirical work.

In addition to these pointwise diagnostics, we can study the *loop area* traced out by trajectories in the $(p_\mathrm{C}, V_\mathrm{C})$ plane over policy or credit cycles. In an idealized reversible limit (no dissipation,

Table 2: Illustrative sign patterns in expansion/contraction regimes. "Expansion with headroom use" corresponds to broad-based credit growth ($\Delta M_{\text{in}} > 0$, $\Delta S_{\text{M}} > 0$) that eats into capacity ($\Delta V_{\text{C}} < 0$). "Expansion with headroom rebuild" captures cases where asset-side growth and recapitalisation outpace use ($\Delta M_{\text{in}} > 0$, $\Delta V_{\text{C}} > 0$). "Contraction with headroom rebuild" refers to deleveraging or freezing of activity while buffers are restored ($\Delta M_{\text{in}} < 0$, $\Delta V_{\text{C}} > 0$). "Contraction with headroom use" flags stressed regimes in which activity shrinks yet capacity is still eroded ($\Delta M_{\text{in}} < 0$, $\Delta V_{\text{C}} < 0$; ΔS_{M} can be ambiguous).

Regime	ΔM_{in}	ΔS_{M}	ΔV_{C}
Expansion with headroom use	$+$	$+$	<0
Expansion with headroom rebuild	$+$	$\gtrsim 0$	>0
Contraction with headroom rebuild	<0	≤ 0	>0
Contraction with headroom use	<0	ambiguous	<0

fixed ξ), the line integral would vanish for closed cycles $\mathcal{C}$, reflecting path-independence on the local equilibrium manifold. In practice, non-zero loop area measures irreversibility and dissipative effects analogous to hysteresis in magnetic systems or frictional cycles. The streaming estimator in the code updates an exponentially weighted version of this integral via

$$W_t = \lambda W_{t-1} + p_{\text{C}}(t-1)\big(V_{\text{C}}(t) - V_{\text{C}}(t-1)\big),$$

with forgetting factor λ chosen so that old cycles gradually lose influence.

Large or persistent values of W_t therefore flag regimes where policy paths in $(p_{\text{C}}, V_{\text{C}})$ space exhibit pronounced hysteresis and stress.

3.3 Dissipation and a credit quality factor Q

In the data, irreversibility appears as non-zero loop area in the $(p_{\text{C}}, V_{\text{C}})$ plane. In physics, a similar idea is captured by the quality factor Q of an oscillator: it compares stored energy with the energy lost in one cycle. Here we define an analogous quality factor for the credit system.

Let loop_area(t) be the dissipative part of the policy or credit cycle in period t and let $X_{\text{C}}(t)$ be the exergy-like "usable credit capacity" at time t. We first define a simple loss ratio

$$\phi(t) = \frac{\text{loop_area}(t)}{X_{\text{C}}(t-1)}, \tag{17}$$

whenever $X_{\text{C}}(t-1) > 0$. When $X_{\text{C}}(t-1) \leq 0$, $\phi(t)$ and $Q_{\text{C}}(t)$ are left undefined (treated as missing) in the baseline implementation. This tells us what fraction of available exergy is lost through policy hysteresis in that period. We then define the credit quality factor

$$Q_{\text{C}}(t) = \frac{2\pi}{\phi(t)} = 2\pi\,\frac{X_{\text{C}}(t-1)}{\text{loop_area}(t)}. \tag{18}$$

A large Q_{C} means that the system loses only a small share of usable credit capacity in each cycle. A small Q_{C} means that even modest policy loops destroy a meaningful part of X_{C}, which is consistent with a fragile or strongly path-dependent credit structure.

For mechanical oscillators, a standard first-order approximation is $\Delta E/E \approx -2\pi/Q$. By analogy, we can write

$$\frac{\Delta X_{\text{C}}(t)}{X_{\text{C}}(t-1)} \approx -\frac{2\pi}{Q_{\text{C}}(t)} + \text{external_injection}(t) + \varepsilon_t, \tag{19}$$

where the external-injection term collects net additions of usable credit capacity (for example, through large-scale asset purchases or recapitalisation episodes). In this way Q_{C} acts as a simple, dimensionless summary of how much hysteresis the credit system can tolerate before its usable capacity is eroded.

Interpreting W_{policy}. In the abstract decomposition (16), W_{policy} collects structured interventions (policy-rate shifts, large-scale asset purchases, regulatory changes) that are not captured by the dispersion and capacity-use terms. In the reference implementation we do not estimate $W_{\mathrm{policy}}(t)$ separately inside the indicator pipeline. Instead we compute the finite-difference terms ΔU, $\bar{T}_L \Delta S_{\mathrm{M}}$, and $-\bar{p}_{\mathrm{C}} \Delta V_{\mathrm{C}}$ and treat the residual $\Delta U - (\bar{T}_L \Delta S_{\mathrm{M}} - \bar{p}_{\mathrm{C}} \Delta V_{\mathrm{C}})$ (the "first-law residual") as a combined bucket for policy work and model error. For case studies or event analyses, users can overlay simple parametric forms—for example, specifying $W_{\mathrm{policy}}(t) = \gamma \mathbf{1}\{t \in [t_0, t_1]\}$ around a major programme such as a large-scale asset purchase episode, or using regression-style dummies—to attribute part of the residual to known policy actions while leaving the remainder as ε.

3.4 Free Energy, Exergy, and Early-Warning Gauges

We seek a scalar gauge that (i) decreases as dispersion rises under a fixed environment, and (ii) upper-bounds structured work extractable over a cycle. A Helmholtz-style free energy provides this.

For a fixed reference environment (T_0), define

$$F_{\mathrm{C}} \equiv U - T_0 S_{\mathrm{M}}. \tag{20}$$

Then, under suitable regularity conditions,

$$\mathrm{d}F_{\mathrm{C}} = -p_{\mathrm{C}}\, \mathrm{d}V_{\mathrm{C}} + \delta W_{\mathrm{other}}, \tag{21}$$

so that F_{C} plays the role of an available-potential measure. As $\Delta F_{\mathrm{C}} \to 0$ under chosen boundaries, policy headroom for structured adjustments is effectively exhausted.

When an ambient pressure-like parameter p_0 is also relevant, an exergy-like functional is

$$X_{\mathrm{C}} = (U - U_0) + p_0(V_{\mathrm{C}} - V_{\mathrm{C},0}) - T_0(S_{\mathrm{M}} - S_{\mathrm{M},0}), \tag{22}$$

which reduces to $-\Delta F_{\mathrm{C}}$ if V_{C} is fixed and p_0 effects are negligible. Because X_{C} depends on boundary choices (T_0, p_0), we treat it as an optional, environment-dependent gauge.

In principle X_{C} can take either sign, depending on the chosen reference state $(U_0, V_{\mathrm{C},0}, S_{\mathrm{M},0})$ and environmental parameters (T_0, p_0). Negative values indicate that the current configuration has lower "available credit energy" than the reference; for early-warning visualisation, however, it is often convenient to enforce a non-negative floor (by clipping or by subtracting the minimum value over the sample) so that zero marks the effective boundary of usable headroom.

For regime diagnostics it is also useful to separate surplus and shortage components relative to a fixed free-energy reference. Let $\Delta F_C(t) = F_C(t) - F_C^{\mathrm{ref}}$ for some chosen baseline F_C^{ref} (e.g. the first non-missing observation or a user-specified date). We then define

$$X_C^+(t) = \max\{0,\ \Delta F_C(t)\}, \qquad X_C^-(t) = \max\{0,\ -\Delta F_C(t)\},$$

so that X_C^+ tracks "excess" free energy above the reference and X_C^- tracks shortfall. These series enter plots and summary tables as complementary early-warning gauges: persistent build-up of X_C^+ suggests unusually large policy room relative to the reference regime, whereas elevated X_C^- points to sustained pressure towards the boundary of feasible adjustment.

A Gibbs-style free energy

$$G_{\mathrm{C}} \equiv U + p_0 V_{\mathrm{C}} - T_0 S_{\mathrm{M}} \tag{23}$$

can be used when (T_0, p_0) are the natural controls. In applications, we typically monitor F_{C} (fixed environment) and compare with X_{C} or G_{C} when capacity constraints are explicit (cf. exergy notions in [11]).

External coupling. In applications we also allow the credit system to couple to an external "pressure bath" and "temperature bath". The coupling is built in two stages. First, we map a small set of market driver series into standardised monthly processes. For JP, the pressure drivers are conceptually:

- the US high-yield option-adjusted spread (HY OAS), used in level form as a proxy for global credit stress;
- the US–JP 10Y yield spread, $\Delta y_{US,JP}(t) = y_{US}^{10\mathrm{Y}}(t) - y_{JP}^{10\mathrm{Y}}(t)$;
- USD/JPY log returns, $r_{\mathrm{FX}}(t) = \log \mathrm{USDJPY}(t) - \log \mathrm{USDJPY}(t - \Delta t)$.

The temperature driver is the VIX equity volatility index, again mapped to monthly frequency. Each raw series $X_k(t)$ is aggregated to month-start, transformed according to its role (level, spread, log-return), and converted to a z-score

$$Z_k(t) = \frac{X_k(t) - \mathbb{E}[X_k]}{\sigma[X_k]},$$

using sample mean and standard deviation over the available history. Second, we form composite indices by simple averaging over the available standardised drivers:

$$E_p(t) = \frac{1}{K_p} \sum_{k \in \mathcal{P}} Z_k(t), \qquad E_T(t) = \frac{1}{K_T} \sum_{k \in \mathcal{T}} Z_k(t),$$

where $\mathcal{P}$ indexes pressure drivers and $\mathcal{T}$ indexes temperature drivers. Missing values in individual Z_k are ignored on each date so long as at least one driver remains. Equal weights reflect an agnostic prior about the relative importance of each driver. With only a handful of series, more elaborate factor models or time-varying loadings would be weakly identified and would risk overfitting. The unweighted mean therefore serves as a transparent first pass. In future work one could replace this averaging with a statistically estimated factor structure or region-specific loadings once longer histories and stronger identification are available.

Given E_p and E_T we work with effective fields

$$p_{\mathrm{C}}^{\mathrm{eff}}(t) = p_{\mathrm{C}}(t) + \alpha E_p(t), \qquad T_{\mathrm{L}}^{\mathrm{eff}}(t) = T_{\mathrm{L}}(t) + \delta E_T(t),$$

so that external stress and risk-on/off conditions act as linear perturbations to the internal equation of state. When α or δ are set to zero the system is "isolated" in this sense; non-zero couplings encode how exposed the credit state is to external financial conditions.

Sectoral "chemical potentials". Given the MECE allocation shares $\{q_i\}$ and M_{in}, we can form sectoral analogues of chemical potentials. Using the normalised shares $\tilde{q}_i$ from the entropy construction, one defines

$$\mu_i = T_0 k M_{\mathrm{in}} (\log \tilde{q}_i + 1),$$

so that, up to constants, μ_i is the derivative of the entropy-like term with respect to the bucket mass. For interpretation we work with deviations from the cross-bucket mean, $\Delta\mu_i = \mu_i - \bar{\mu}$, where $\bar{\mu}$ is the average over i at each date. Large positive $\Delta\mu_i$ flag categories where incremental "credit mass" is unusually valuable in entropy terms; large negative values flag categories where allocations are already heavy relative to the rest of the system. In the current implementation these μ_i and $\Delta\mu_i$ are used as diagnostic series and are not themselves fed back into the flow dynamics.

3.5 Maxwell-Like Relation, Exterior Derivative, and Legendre Structure

In the local-equilibrium limit where $U(S_\mathrm{M}, V_\mathrm{C}, \xi)$ is approximately state-like with ξ effectively fixed over the sampling window, mixed partial derivatives commute and we obtain a Maxwell-like condition:

$$\left(\frac{\partial T_\mathrm{L}}{\partial V_\mathrm{C}}\right)_{S_\mathrm{M},\xi} = -\left(\frac{\partial p_\mathrm{C}}{\partial S_\mathrm{M}}\right)_{V_\mathrm{C},\xi}. \tag{24}$$

Outside this limit, the failure of integrability is informative rather than a fatal inconsistency.

Geometrically, the pair $(T_\mathrm{L}, p_\mathrm{C})$ defines a differential one-form on the observed $(S_\mathrm{M}, V_\mathrm{C})$ state plane,

$$\omega \equiv T_\mathrm{L}\,\mathrm{d}S_\mathrm{M} - p_\mathrm{C}\,\mathrm{d}V_\mathrm{C}. \tag{25}$$

In the integrable (state-like) regime this one-form is (approximately) exact and can be written as $\omega \approx \mathrm{d}U_\mathrm{rev}$ on a local-equilibrium manifold where ξ is effectively constant over the window. Outside this regime, ω should be treated as a convenient field representation rather than as the differential of a globally defined potential.

Exactness is equivalent to vanishing exterior derivative $\mathrm{d}\omega = 0$ (no "curl"), which in coordinates yields the Maxwell-like condition (24). To quantify departures from integrability, we introduce a non-equilibrium "vorticity" (curl) measure

$$\Omega(S_\mathrm{M}, V_\mathrm{C}) \equiv \left(\frac{\partial T_\mathrm{L}}{\partial V_\mathrm{C}}\right)_{S_\mathrm{M}} + \left(\frac{\partial p_\mathrm{C}}{\partial S_\mathrm{M}}\right)_{V_\mathrm{C}}. \tag{26}$$

Thus $\Omega \approx 0$ corresponds to the integrable limit (local state-like behaviour with fixed ξ), whereas persistent large $|\Omega|$ indicates strong path dependence / structural distortions and motivates modelling ξ explicitly rather than forcing a single-potential representation.

This also clarifies the Legendre structure. Starting from $U(S_\mathrm{M}, V_\mathrm{C})$ we can, in principle, define a potential written in terms of the intensive variables,

$$\Phi(T_\mathrm{L}, p_\mathrm{C}) \equiv U(S_\mathrm{M}, V_\mathrm{C}) - T_\mathrm{L} S_\mathrm{M} + p_\mathrm{C} V_\mathrm{C},$$

where $(S_\mathrm{M}, V_\mathrm{C})$ are now regarded as functions of $(T_\mathrm{L}, p_\mathrm{C})$ via the equation of state. Taking the differential and using (6) one finds

$$\mathrm{d}\Phi = -S_\mathrm{M}\,\mathrm{d}T_\mathrm{L} + V_\mathrm{C}\,\mathrm{d}p_\mathrm{C},$$

so that $(\partial\Phi/\partial T_\mathrm{L})_{p_\mathrm{C}} = -S_\mathrm{M}$ and $(\partial\Phi/\partial p_\mathrm{C})_{T_\mathrm{L}} = V_\mathrm{C}$. This Legendre transform from $(S_\mathrm{M}, V_\mathrm{C})$ to $(T_\mathrm{L}, p_\mathrm{C})$ is locally well-defined only when (24) holds on an approximately integrable manifold. The free-energy-type potentials introduced in §5, such as F_C and G_C, can then be viewed as partially Legendre-transformed versions of U evaluated at fixed environmental parameters (T_0, p_0).

Empirically, one can estimate $T_\mathrm{L}(S_\mathrm{M}, V_\mathrm{C})$ and $p_\mathrm{C}(S_\mathrm{M}, V_\mathrm{C})$ from proxy series and test whether (24) approximately holds. Because U, T_L, and p_C are built from partly independent proxies rather than from a single fitted potential, the relation is not enforced by construction. Instead, it works as an integrability diagnostic. Persistent, large violations suggest non-equilibrium behaviour, changing ξ, or proxy misspecification. In this sense, the analogy is falsifiable rather than purely rhetorical, and Section 4 summarizes a simple set of operational diagnostics built from this condition.

4 Practical Tests and Falsifiability

The mapping yields several operational diagnostics:

1. **Integrability / vorticity test.** Estimate local derivatives in (24) and compute the vorticity Ω in (26). Large, persistent $|\Omega|$ is interpreted as a non-equilibrium signal (path dependence / structural distortions), not merely "model failure".

2. **Work vs. dispersion decomposition.** Use (16) to decompose changes in U into heat-like dispersion ($\bar{T}_L \Delta S_{\mathrm{M}}$), capacity use ($-\bar{p}_{\mathrm{C}} \Delta V_{\mathrm{C}}$), and policy work ($W_{\mathrm{policy}}$).

3. **Free-energy and exergy ceilings.** Monitor F_{C} or X_{C} as early-warning gauges: sustained proximity to zero under stress scenarios flags limited room for non-disruptive adjustment.

4. **Loop area and hysteresis.** Compute loop areas in $(p_{\mathrm{C}}, V_{\mathrm{C}})$ over policy cycles. Non-zero areas capture dissipative stress and irreversibility in credit allocation dynamics[K17].

5. **Entropy production proxy.** Use loop area, $\mathcal{D}$, and changes in Q_{C} as operational proxies for the entropy production rate σ_S in (7). Rising dissipation and falling Q_{C} serve as early-warning signals of irreversible stress build-up.

All of these rely on observable or reconstructible quantities from public balance-sheet and market data. Implementation details such as proxy choices, normalisation, and robustness checks remain part of ongoing empirical work and should be reported alongside any results.

4.1 Testable hypotheses

The reduced-form version of Thermo Credit should be judged by a short list of predictive hypotheses rather than by conceptual elegance alone:

- **H1.** Lower q_t should predict stronger asset-price acceleration, especially in land, housing, and other existing-asset markets, before it predicts stronger real growth.

- **H2.** Higher C_t with lower q_t should show weaker pass-through to inflation and GDP-linked activity than equally large credit expansions with higher q_t.

- **H3.** High C_t^A combined with high S_t should predict greater future volatility, spread widening, or contraction risk.

This yields explicit rejection criteria. In the measurement-first reading of the model, the framework is weakened if:

- separating C_t^R and C_t^A does not improve forecasting or regime classification relative to total-credit-only baselines;

- estimated declines in q_t fail to line up with later asset acceleration or fragility build-up;

- adding the stress state S_t does not improve downside prediction beyond the existing thermo-monitoring indicators.

4.2 Implicit state relation and iterative calibration

At this point the practical weakness is clear. If $(T_0, p_0, U_0, V_0, S_0)$ are left as fixed convenience constants, then X_C is only a transformation of existing series and can easily lose economic meaning. The operational version therefore treats the state as *implicitly* defined by observables and a region-specific parameter vector

$$\theta_r \equiv (T_{0,r}, p_{0,r}, U_{0,r}, V_{0,r}, S_{0,r}).$$

The minimum implicit system is

$$0 = g_{1,t}(\theta_r) \equiv X_C^{\theta_r}(t) - \Big[(U_t - U_{0,r}) + p_{0,r}(V_{C,t} - V_{0,r}) - T_{0,r}(S_{M,t} - S_{0,r})\Big], \tag{27}$$

$$0 = g_{2,t}(\theta_r) \equiv \eta_t - \eta_0 + \alpha(1 - q_t) + \beta S_{t-1}, \tag{28}$$

$$0 = g_{3,t}(\theta_r) \equiv S_t - \rho S_{t-1} - \gamma C_t^A / Y_t^N + \delta C_t^R / Y_t^N. \tag{29}$$

Equations (27)–(29) are not a full structural model. They are a practical identification layer. They say that the headline variables only become useful when the reference parameters and the latent states are tied back to data.

In plain language, this block does three jobs. Equation (27) says that headroom is not observed directly. We infer it from output, capacity, and entropy after we choose a regional reference point. Equation (28) says that efficiency should fall when existing-asset credit and stress become too strong. Equation (29) says that fragility should rise when existing-asset credit keeps building up, and should ease when credit moves back to GDP-linked uses.

For the current repository implementation we therefore use a lightweight, iterative calibration step rather than a one-time manual choice. For each region r, the working objective is

$$\begin{aligned}\mathcal{L}_r(\theta_r) = {} & w_g\Big[1 - \mathrm{corr}\big(X_C^{\theta_r}(t), \Delta U_{t+h}\big)\Big] + w_s\Big[1 + \mathrm{corr}\big(X_C^{\theta_r}(t), \Delta \mathrm{spread}_{t+h}\big)\Big] \\ & + w_\ell\Big[1 + \mathrm{corr}\big(X_C^{\theta_r}(t), |\mathrm{loop_area}_{t+h}|\big)\Big] + w_T\Big[1 - \mathrm{corr}\big(X_C^{\theta_r}(t), T_{L,t}\big)\Big] + w_R\|\theta_r - \theta_r^{(0)}\|^2,\end{aligned} \tag{30}$$

with horizon $h = 4$ quarters in the current empirical pass and $\theta_r^{(0)}$ denoting the configuration defaults. In simple terms, a useful X_C should move with later capacity growth, move against later spread widening and dissipation, and still stay close enough to the default scale to remain readable. The present code implements this as a small coordinate search, not as a claim of final identification. That is deliberate: the immediate goal is to stop X_C from becoming a decorative transform and to make the parameter choice open to testing.

In the implementation, the level anchors (U_0, V_0, S_0) are first re-centered on each region's own sample medians before the iterative search begins. This is important in practice. Public regional panels often mix quantities with very different nominal scales, and the raw configuration constants can otherwise make X_C look like a unit-conversion artifact. The tuned score should therefore be read as an *implicit headroom index*: it keeps the sign logic of X_C, but it is identified from local deviations around observed regional baselines rather than from fixed absolute constants.

This also gives a simple reading rule for the results section. If the tuned implicit headroom score moves while the raw pipeline X_C stays flat or clipped, the original fixed-parameter transform is too rigid for that region. If the tuned score still fails to improve the relation to future growth or stress, then the thermo-style layer is not adding enough value.

5 Limits, Identification, and Scope

This construction is intentionally modest. Key limitations include:

- **Category dependence.** S_{M} depends on the chosen partition of uses. Results should be checked across other reasonable partitions.
- **Proxy noise.** T_{L}, V_{C}, p_{C} are built from noisy proxies, and U itself is instantiated from GDP- or credit-like aggregates. Measurement error or proxy misspecification can break (24) and distort the decomposition (16); in practice these relations are treated as reference structures and consistency checks rather than hard identities.
- **Local-equilibrium vs. non-equilibrium.** Fast crises and regime shifts violate smooth state-variable assumptions. We therefore interpret the Maxwell-like relations as local limit cases and treat vorticity and hysteresis measures as the main non-equilibrium outputs.
- **Non-physical.** We do not assume microscopic "money particles" or claim that macro-financial data obey physical first/ second laws. The mapping is an analogy and is judged only by empirical usefulness.
- **Identification challenges.** Policy, expectations, and shocks act jointly. Causal claims need careful research design beyond the bookkeeping framework presented here.
- **Partition measurement.** The coarse split (C_t^R, C_t^A) is often only partially observed in public data. In many applications, q_t must be proxy-estimated or treated as a latent state, so reported results should separate direct observations from inferred quantities. This matters especially for construction credit, which often sits between GDP-linked and existing-asset uses. In practice the weighting parameter λ_B is itself part of identification rather than a universal constant.
- **Scaling conventions.** Constants such as k are conventional. Report normalized metrics and sensitivity checks.

This version should therefore be read as a theory-and-implementation note with explicit rejection conditions. If the partition variables (C_t^R, C_t^A, q_t) and the stress state S_t fail to improve empirical discrimination relative to simpler credit-volume baselines, the measurement-first version of Thermo Credit should be simplified or rejected in that application. A more systematic empirical evaluation for the JP configuration is left to a later revision.

6 Data Sources and Reproducibility

6.1 Public data sources

The current build draws on public macro-financial sources and stores harmonised tables in the repository. Credit stocks come from BIS private non-financial credit series and the region-specific credit tables under `data/`. Nominal output proxies are taken from World Bank caches and, for the euro area when available, ECB quarterly GDP. Money and balance-sheet scale proxies are selected from central-bank balance sheets or broad-money series (BoJ, ECB, Fed) through the region builders. Market-stress inputs include long-term government yields, ICE BofA high-yield spreads, VIX, and selected foreign-exchange and yield-spread series used in the external-coupling block. The repo-level mapping of symbols, status, and construction rules is recorded in `data/data_dictionary.csv`.

6.2 Empirical snapshot from the current build

The theory note now reads back a small empirical summary from the same indicator tables used by the dashboard, so that the text does not stop at definitions alone. The aim of the snapshot is narrow but useful: it reports where the latest regional coverage ends, whether the tuned implicit headroom signal says anything that the raw pipeline X_C does not, and whether the current data support a directional read on stress and dissipation.

Latest cross-region snapshot.

- Coverage in the current theory build: Japan (JP) through 2025-03-31, Euro Area (EU) through 2025-09-30, United States (US) through 2025-03-31.
- Japan (JP): the calibrated implicit headroom score is 1.994, pipeline X_C is 1,090,132.075, latest loop area is 2,808,932, and T_L is currently high in its own regional history; implicit headroom is rising over the last four quarters; loop dissipation is easing over the last four quarters; forward-growth correlation improves from -0.747 to 0.495.
- Euro Area (EU): the calibrated implicit headroom score is 1.938, pipeline X_C is 0.000, latest loop area is 106, and T_L is currently high in its own regional history; implicit headroom is roughly flat over the last four quarters; loop dissipation is worsening over the last four quarters; forward-growth correlation improves from 0.000 to 0.269; pipeline X_C is clipped at zero at the sample end, so calibration is doing real work.
- United States (US): the calibrated implicit headroom score is 1.477, pipeline X_C is 3,560,611.529, latest loop area is 0, and T_L is currently high in its own regional history; implicit headroom is falling over the last four quarters; loop dissipation is unclear over the last four quarters; forward-growth correlation does not yet improve over pipeline X_C (-0.076 versus 0.113).

6.3 Illustrative calibration output

The current repository pass also emits a lightweight calibration table for the region-specific reference parameters. This is not yet a final estimator, but it makes the tuning step visible and reproducible. In particular, the table should be read together with the snapshot above: the important question is not whether the fitted constants look elegant, but whether the tuned signal improves the forward-looking relation to growth and stress relative to the raw dashboard transform.

Table 3: Illustrative region-by-region calibration of $(T_0, p_0, U_0, V_0, S_0)$.

Region	End date	T_0	p_0	U_0	V_0	S_0	Corr. to ΔU_{t+4}	Corr. to $\Delta\text{spread}_{t+4}$
Japan (JP)	2025-03-31	0.293	1.707	809,879	1,072,698	2,313,837	0.495	0.381
Euro Area (EU)	2025-09-30	1.043	1.618	2,052,594	50	161	0.269	0.178
United States (US)	2025-03-31	1.087	1.855	21,585	1,891,556	6,059,079	-0.076	0.024

The search is deliberately lightweight. Level anchors are first centered on each region's sample medians, so the tuned score should be read as an implicit local-headroom index rather than as a raw nominal level.

6.4 Paper-ready figures

For the narrative note itself, the repository now also generates a small set of paper-ready vector figures directly from the same indicator CSVs used by the dashboard. These figures are exported as

PDF/SVG via a `matplotlib/seaborn` pipeline so that the paper layer does not depend on browser screenshots or Plotly-specific rasterization.

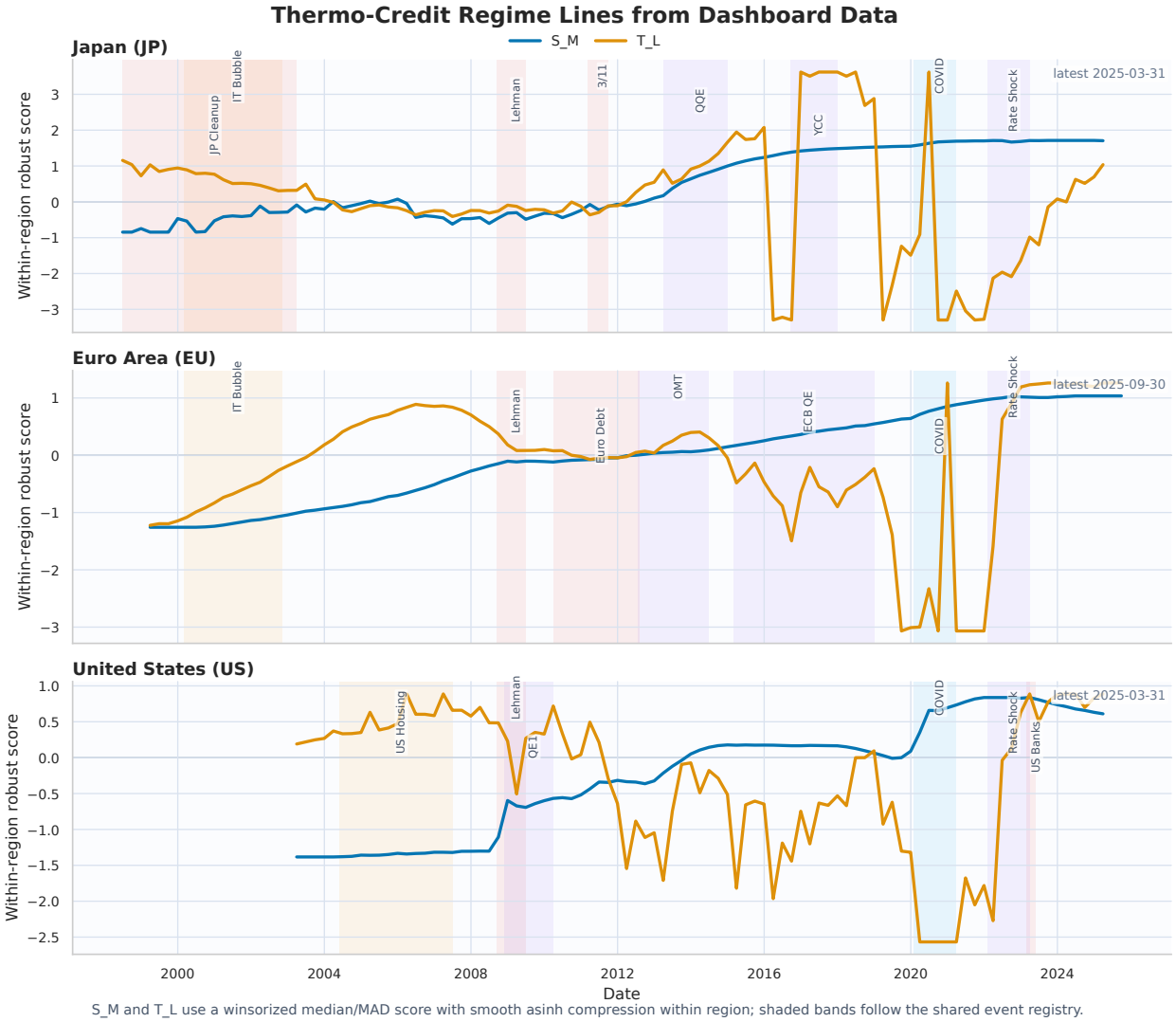

Figure 1: Regional panels for S_M and T_L, rebuilt from the dashboard indicator tables. Lines are shown as winsorized median/MAD scores with smooth `asinh` compression, and shaded windows reuse `data/report_events.csv`. The build may use a fresher git ref when a feature branch carries stale regional CSVs.

6.5 Reproducibility

This note is designed to accompany an openly available codebase and data snapshots such as monthly reports computed from public sources. The reference implementation lives in the repository `2025_11_Thermo_Credit` and is intended to make all reported figures and indicators reproducible.

At a high level:

- a Python pipeline computes S_{M}, T_{L}, V_{C}, F_{C}, X_{C}, and the diagnostics from public time series using shared configuration files and scripts;

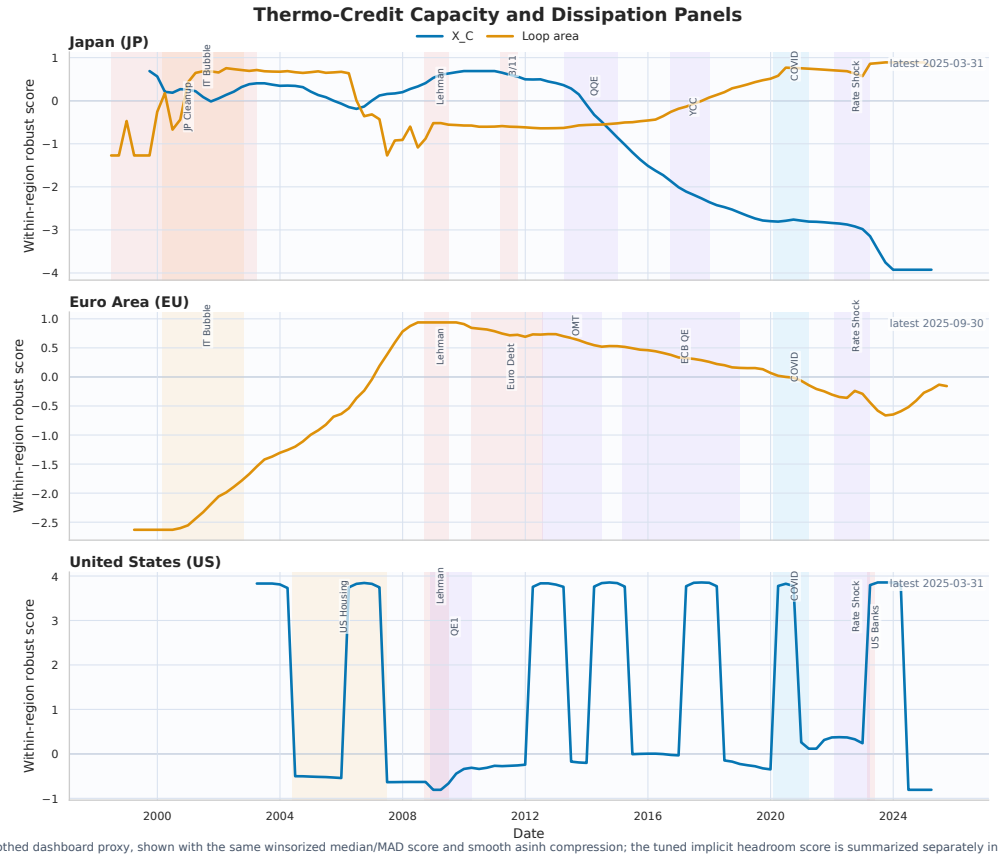

Figure 2: Regional panels for X_C and loop dissipation, rebuilt from the dashboard indicator tables. For X_C, the paper version applies a trailing four-quarter rolling median and a light two-quarter mean before the same winsorized median/MAD scaling and event overlay used elsewhere. This panel remains a visual proxy from the dashboard build; the tuned implicit headroom score discussed in the text is reported in the empirical snapshot and calibration table below.

- minimal synthetic CSVs are provided so that tests and examples can run without external API keys, while real data are fetched from standard sources (e.g. FRED, World Bank) when credentials are present;
- continuous-integration workflows (GitHub Actions) pin Python dependencies and re-run the pipeline so that indicators and the report can be regenerated from a fixed environment.

Further implementation details (module structure, configuration roles, CI layout, data-source routing, and synthetic-data conventions are summarised in Appendix A.

Drafting support from a GPT-based large language model was used as an editing tool. All equations, definitions, and claims remain the responsibility of the authors.

Disclaimer. This material is part of an ongoing research program. It is provided "as is" without any warranty of accuracy, completeness, or fitness for a particular purpose. It does not constitute investment, legal, tax, or policy advice, and does not create a client relationship or any regulated financial service. Users are responsible for compliance with applicable laws in their jurisdiction.

Appendix A: Data Sources and Implementation Notes

This appendix summarizes the main implementation modules for readers who wish to reproduce or extend the reference pipeline. It is not required for reading the theoretical sections.

Data-source map. The region builders write harmonised intermediate tables under `data/`. In the current repository, Japan mainly uses BoJ balance-sheet and yield proxies together with World Bank GDP caches; the euro area uses ECB or FRED/World-Bank mirrors for money, GDP, and yields, plus BIS credit series; and the United States uses Federal Reserve balance-sheet or M2 proxies, US Treasury yields, BIS credit, and the same World Bank GDP layer. Cross-market stress inputs used in the external-coupling block include ICE BofA high-yield spreads, VIX, and selected FX / yield-spread series. The exact symbol-to-file mapping is stored in `data/data_dictionary.csv`.

- **Raw and feature tables.** Region-specific builder scripts `scripts/01_build_features.py` (JP), `scripts/04_build_features_eu.py` (EU), and `scripts/05_build_features_us.py` (US) select source series according to configuration files (`config.yml`, `config_{jp,eu,us}.yml`) and `lib/series_selector.py`. They write money, credit, regulatory-pressure, and allocation tables under `data/` (e.g. `money*.csv`, `credit*.csv`, `reg_pressure*.csv`, `allocation_q*.csv`).

- **Entropy and temperature.** The module `lib/entropy.py` implements $S_{\mathrm{M}} = kM_{\mathrm{in}}H(q)$ from (4) and also computes $S_{\mathrm{M,in}}$, $S_{\mathrm{M,out}}$, $S_{\mathrm{M,hat}} = H(q)/\log K$, and optional per-bucket contributions. The module `lib/temperature.py` builds T_{L} from credit spreads, depth, and turnover, combines the standardised inputs into $\hat{T} = z_s z_d(1 + \frac{1}{2}z_t)$, and then applies min–max normalisation.

- **Internal-energy gauge.** In the credit tables, the column $U(t)$ acts as the internal-energy or capacity proxy used in (16), (20), and (22). When a dedicated U series is missing, the code fills it conservatively: first from an explicit U column, then from GDP-like series such as $U_{\mathrm{gdp_only}}$ or Y, and only as a last resort from L_{real}. An optional detrended version $U^{\star}$ subtracts a no-lookahead rolling mean.

- **Depth and turnover enrichment.** When region-specific depth/turnover series are not supplied, the helper `lib/credit_enrichment.py` constructs depth and turnover heuristically from real credit stocks L_{real} and capacity proxy U as described in the main text.

- **Measurement-first tool layer.** The module `lib/thermo_credit_tools.py` provides the compact v2 interface used by the repository-side AI layer: theory overview, variable definitions, metric computation, scenario evaluation, and regime comparison. The script `scripts/thermo_credit_cli.py` exposes the same layer as a JSON CLI for local checks.

- **MCP distribution wrapper.** The module `lib/thermo_credit_mcp.py` wraps the same core calculations as MCP resources, tools, and prompt templates, and `scripts/thermo_credit_mcp_server.py` provides a runnable server entrypoint. This layer is meant for AI accessibility and structured reuse. It does not replace the empirical validation logic in the main text.

- **Capacity, headroom, and coupling.** The core constructor `lib/indicators.py` merges money, allocation, credit, and regulatory tables, renames regulatory quantities into V_{C} and p_{C}, and, when configured with the "min_headroom" formula, sets $V_{\mathrm{C}}(t) = \min_j H_j(t)$ over headroom components. External pressure/temperature drivers are aggregated in `lib/external_coupling.py` into E_p and E_T; when the coupling coefficients α, δ are non-zero

the code applies the affine shifts $p_{\mathrm{C}} \leftarrow p_{\mathrm{C}}^{\mathrm{baseline}} + \alpha E_p$ and $T_{\mathrm{L}} \leftarrow T_{\mathrm{L}}^{\mathrm{baseline}} + \delta E_T$. The same module computes F_{C}, X_{C}, and the Gibbs-style G_{C} and streams the loop-area estimator in $(p_{\mathrm{C}}, V_{\mathrm{C}})$.

- **Diagnostics.** The diagnostics routine in `lib/indicators.py` estimates $\partial S_{\mathrm{M}}/\partial V_{\mathrm{C}}|_{T_{\mathrm{L}}}$ and $\partial p_{\mathrm{C}}/\partial T_{\mathrm{L}}|_{V_{\mathrm{C}}}$ via rolling least squares, and constructs finite-difference versions of the first-law decomposition by computing ΔU, ΔS_{M}, ΔV_{C}, $\bar{T}_{\mathrm{L}}$, $\bar{p}_{\mathrm{C}}$, and the implied Q_{like} and W_{like} series.

- **Per-region orchestration.** Scripts `scripts/02_compute_indicators.py`, `scripts/02_compute_indicators_eu.py`, `scripts/02_compute_indicators_us.py`, and the helper `lib/regions.py` assemble the inputs for JP/EU/US and write `site/indicators*.csv` used by the dashboard and tests.

- **Theory calibration layer.** The module `lib/theory_calibration.py` and script `scripts/07_calibrate_t` perform a lightweight iterative search over $(T_0, p_0, U_0, V_0, S_0)$, write `data/calibrated_theory_params.json`, and emit the LaTeX snippets consumed in the empirical snapshot and calibration sections of the note.

- **Fallbacks and CI reproducibility.** To keep tests and scheduled builds deterministic, `scripts/ci_prepare_minimal_data.py` seeds tiny synthetic CSVs when real inputs are absent, and the GitHub Actions workflows re-run the pipeline under pinned dependencies so that indicators and the report can be regenerated from a fixed environment.

Appendix B: Glossary (For Physicists New to Finance)

Numbering. Each term is assigned a unique K-number (K1, K2, ...) in the order it is first referenced in the main text. Use the superscript K# markers to jump back here; headings show the same inline K#.

This glossary explains the main banking, credit, and regulatory terms used in this note for readers with a physics background. The definitions are informal and written for intuition. Where useful, we add analogies to thermodynamics or dynamical systems. These analogies are only guides, not literal identities.

1. Regulation and Bank Balance Sheets

K1 CET1 (Common Equity Tier 1). This is the core equity capital of a bank, such as common shares and retained earnings. It absorbs losses and supports solvency. Analogy: the bank's basic "energy reserve".

K2 RWA (Risk-Weighted Assets). These are total assets weighted by regulatory risk factors. Riskier exposures get higher weights. The ratio CET1/RWA is one of the main prudential metrics. Analogy: an effective load or mass adjusted for fragility.

K3 CET1/RWA ratio and slack. The CET1/RWA ratio shows how much high-quality capital backs risk-weighted assets. Regulation sets a minimum. The slack is the amount above that minimum, and it measures capacity to expand or absorb shocks. Analogy: a safety margin or unused headroom in a constrained system.

K4 Credit capacity / Headroom (V_{C}). This is the effective room for additional lending, given capital, liquidity, and internal limits. In this note, V_{C} is the state-like variable that captures that remaining room. Analogy: usable volume before the system hits a hard wall.

K5 Credit pressure (p_{C}). This is the shadow price of capacity: the marginal cost or value of relaxing or tightening V_{C}. It is defined by analogy to $p_{\mathrm{C}} = -(\partial U/\partial V_{\mathrm{C}})_{S_{\mathrm{M}}}$ or $-\partial F/\partial V_{\mathrm{C}}$. Analogy: thermodynamic pressure; tighter constraints imply higher p_{C}.

K6 HQLA (High Quality Liquid Assets). These are very liquid, low-risk assets, such as government bonds, that can be sold quickly in stress. They support liquidity ratios and crisis resilience. Analogy: high-grade stored energy that can be tapped on demand.

K7 Liquidity buffer. This is cash plus HQLA held as an emergency reserve against sudden outflows or market stress. Analogy: a backup battery or safety reservoir.

K8 LCR (Liquidity Coverage Ratio). This regulatory ratio is HQLA divided by projected net cash outflows over a 30-day stress scenario. It tests short-term liquidity robustness. Analogy: can the system keep running for 30 days under worst-case load?

K9 NSFR (Net Stable Funding Ratio). This regulatory ratio looks over a one-year horizon. It compares stable funding with required stable funding and limits excessive maturity mismatch. Analogy: making sure long-lived assets are backed by sufficiently slow-decaying sources.

2. Money, Credit, and Flow Concepts

K10 Money-in-circulation. This is an operational measure of money that is actually circulating in the chosen system and horizon, for example deposits used in payments. It is not just base money on the central bank balance sheet. Analogy: active particles taking part in interactions.

K11 Credit stocks and flows. Stocks are the outstanding amounts of loans or credit at a given time. Flows are the changes per period, such as new lending, repayments, or net issuance. Analogy: internal energy versus power or energy flux.

K12 Margin credit / Securities financing. This is credit used to finance positions in securities, such as margin loans or some repo activity. It amplifies leverage in financial markets and is often separated from credit to the real economy.

K13 Commercial Paper (CP). This is short-term unsecured debt issued by firms. It provides funding outside traditional bank loans. Analogy: an alternative branch in the credit circuit.

K14 Policy work (W_{policy}). This covers deliberate interventions by central banks, regulators, or governments: rate changes, asset purchases, guarantees, capital-rule changes, and similar actions. In our decomposition, this is the structured and intentional part of "work". Analogy: externally applied work on the system.

3. Structural and Analytical Concepts

K15 MECE (Mutually Exclusive, Collectively Exhaustive). This is a rule for defining categories: no overlaps and no gaps. For the entropy $S_{\text{M}} = kM_{\text{in}}H(q)$, the allocation shares q_i should be MECE. Otherwise S_{M} is distorted by double counting or omissions.

K16 State variable and proxy. A state variable is determined by the current state, not by the full path that led there. In practice, finance uses observable proxies such as ratios and indices to approximate such variables. Our Maxwell-like relation (24) tests whether the chosen proxies behave as if they came from a consistent potential $U(S_{\text{M}}, V_{\text{C}}, \ldots)$.

K17 Hysteresis and loop area. If trajectories in the $(p_{\text{C}}, V_{\text{C}})$ plane form loops with non-zero area over policy or credit cycles, this suggests irreversibility or dissipation-like effects, such as stress, misallocation, or path dependence. Analogy: magnetic hysteresis, frictional cycles, or inelastic processes.

K18 Stress testing and early-warning indicators. Stress tests simulate severe but plausible scenarios to see whether banks or systems can survive them. Early-warning indicators try to flag fragility in advance. Within this framework, quantities like F_{C}, X_{C}, and loop areas are meant as structured candidates to complement such tools.

K19 Money vs. credit (QTM vs. QTC perspective). In traditional Quantity Theory of Money (QTM), money is primary and credit is often treated as derivative. In the Quantity Theory of Credit (QTC), and in this note, credit creation is primary: bank lending creates deposits, and money is an accounting outcome of credit decisions. This shift motivates tracking credit capacity and dispersion as state-like objects.

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